

# Verity

Pre-payment fraud tripwire for Medicare and Medicaid, built on public federal data. DNHacks 2026, Health and Public Service track. Summary of all work as of Saturday, September 5, 2026, ~1:00 PM ET. Submissions are due Sunday at noon; demos at 1:00 PM; finals at 4:00 PM.

## 1. What Verity is

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Verity finds providers that should not be paid before the next check goes out, and hands a human reviewer an explainable, auditable packet. It joins a dozen federal datasets nobody joins in one place and runs three detectors over them:

- **Ghost networks.** A graph of hospice, home health, SNF and hospital enrollments, their owners (CMS All Owners files), addresses and phones (NPDES), changes of ownership, and the revoked and excluded lists. Community detection surfaces clusters of newly incorporated entities sharing owners and addresses, benchmarked against CMS Market Saturation providers-per-beneficiary by county.
- **Impossible days.** The 238M-row T-MSIS Medicaid provider spending file (NPI by HCPCS by month, 2018 to 2024, all states), converted through a 318-code table of minutes-per-unit into implied clinician hours per day. Flags providers billing more hours than exist, with robust z-scores by state and code. This is the fraud pattern the White House keeps citing (Minnesota autism and housing programs; DOJ's June 2026 takedown charged "500+ hours of counseling per day").
- **Revoked but paid.** Providers revoked in Medicare, excluded by OIG, debarred in SAM.gov (including OPM/FEHB), listed on state exclusion lists, deactivated in NPDES, or terminated as deceased in Medicaid, who still show Medicaid payments after the event date. OIG found 40 states paid \$50.3M to 584 terminated providers; OBBA forces every state to build exactly this check by December 31, 2026.

On top sits an investigator agent (Codex credits) that assembles the evidence for a flagged cluster, drafts a CMS-style referral packet citing 42 CFR 424.535 grounds, and records the reviewer's accept or reject decision as feedback. The console is a custom Next.js app: light neutral palette, serif display headings, rounded cards, a D3 county map and an ownership force graph.

## 2. Why this idea for this room

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The David Network is a conservative network of ~5,000 students from Ivy-plus schools; DNHacks is its inaugural hackathon and is explicitly trying to redirect elite technical talent toward defense, industry and government. The site's judging language is "technical quality" plus "potential to reduce costs for taxpayers and improve the experience of interacting with government," and the FAQ says judges "will heavily weight the technical depth." First prize is \$5K plus a \$20K grant conditional on continued work, plus a year-round talent cohort and investor meetings. They are selecting people as much as projects.

Reading every judge's public record, the room converges on eight things, and Verity is designed to hit all eight: a sourced dollar figure; government data nobody has joined; visible technical depth; "detect and deploy, not pay and chase"; a human in the loop with explainable flags; agentic AI doing mission work with a feedback loop rather than a chat wrapper; something legacy that gets turned off; and a commercial buyer that is not only the federal government.

### 3. The judges and what each rewards

| Who                                                                                                                 | What the public record says they reward                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
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| <b>Andrew McCarthy</b> , CTO, White House Office of Anti-Fraud Initiatives                                          | Career IC analyst (USAF, NGA, NRO, NIC), CISA chief of staff, then CTO of the office executing EO 14395 (March 2026, Task Force to Eliminate Fraud chaired by VP Vance). Mandate: 30/60/90-day fraud vulnerability reports, pre-payment controls, identity and eligibility verification, federal-state data sharing, disrupting fraud networks. Wins touted: 70 LA hospices suspended in a week via an internal AI platform, \$6.3B in contracts to shell-address firms flagged, \$260M Minnesota Medicaid frozen. No public quotes; judge him by the EO. Punishes sloppy data handling and missing audit trails. |
| <b>Wesley Robbins</b> , Head of ML, Clearview AI (track sponsor)                                                    | Only judge with a real ML publication record: FarSight (IARPA BRIAR, NIST-evaluated), DaliID (recognition robust to badly degraded inputs), CAST (attribute-conditioned evaluation). Clearview is pursuing HHS benefits-fraud pilots and lists "financial investigations" on its federal page, which is why a face-recognition firm sponsors a health track. Rewards rigorous evaluation, held-out tests, ablations, honest error analysis, robustness. Punishes cherry-picked metrics.                                                                                                                           |
| <b>Justin Fanelli</b> , CTO, Department of the Navy                                                                 | Former Chief Data Architect for Defense Health (your direct bridge). Runs literal ROI contests. Quotes: "We're looking for 50% or greater improvement." "Coming with an offering that doesn't divest anything is the slowest way." "If you can't talk about it in terms of what impact it's going to have on people, it's going to get lost." "Humility was the number one trait." Wants a measured baseline, a measured delta, and a legacy process turned off. Was Acting DON CTO, never DoD CIO.                                                                                                               |
| <b>Joseph Larson</b> , VP and Head of Government, OpenAI (keynote); Lee Dunn, Natalie Staudacher, Yiren Lu (OpenAI) | Ex-Anduril, first Deputy CDAO for Algorithmic Warfare, Palantir. "You can't just slap these into sensitive public sector domains." Government "can't deploy these tools without meaningful feedback loops." Wants mission-specific agentic workflows on real data infrastructure, not chat wrappers. Likely decides Best Use of AI.                                                                                                                                                                                                                                                                               |
| <b>Jonathan Merril, MD</b> , CEO, Oncovera (sponsor)                                                                | GWU MD, Harvard/MGH informatics fellow, 30-year health-tech founder (Astute Technology sold to Echo360; VR physical therapy). Lens: access at scale, measurable outcomes, patient-facing clarity, business-model realism. The clinician-founder in the room; the project must survive "how does this work in a clinic."                                                                                                                                                                                                                                                                                           |
| <b>Lorenzo Rizzotti</b> , Neuralink                                                                                 | 21-year-old Rust and systems engineer. Expect "does the code actually run." No public statements.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Gurpreet Chandhoke</b> , Co-Founder, Roman AI (sponsor)                                                          | Inference-infrastructure startup whose site is a manifesto on unit economics ("utilisation as the product"). Likely ex-Deutsche Bank tech banker, Wharton MBA. Rewards cost-per-unit thinking and provenance of numbers.                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Trent Gahm</b> , Principal, DTX Ventures (Open track sponsor)                                                    | DTX was founded by two ex-Palantir operators (one built Palantir's healthcare division). 2026 letter: "The United States doesn't need more science projects. We need companies that prioritize deployment over endless discovery." "We try to avoid businesses that sell exclusively to the government." Avoids "model wrappers." Dislikes political theater.                                                                                                                                                                                                                                                     |
| <b>Brian Murray</b> , Craft Ventures; <b>Liam Corrigan</b> , Sequoia; <b>Matt Cronin</b> , a16z                     | Murray: SaaS metrics, "competitors don't come up enough." Corrigan: physics, hardware, persistence. Cronin: ex-DOJ cybercrime prosecutor, ex-ONCD, wants commercial solutions and acquisition reform, has a fraud prosecutor's brain.                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Second Front</b> (Sweatt, Bosquez, Utt); <b>Tim Booher</b> ,                                                     | Deployability and production over demos (2F); security-first and cost-aware inference (Booher); "what government dataset did you unlock" (Levine's Data Crunch thesis); audit                                                                                                                                                                                                                                                                                                                                                                                                                                     |

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| Dept. of War; <b>FAI</b> (Dauber, Levine); <b>Jake Fischer</b> , Army FM&C; <b>Jeff Berkowitz</b> , Delve; Aslan, DEALSAGE, Even, Narion | readiness and waste reduction (Fischer is Principal Deputy ASA, not the Assistant Secretary); "amplify not replace," precision and transparency (Berkowitz); "scaling existing human judgment" (Aslan). |
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Unverified: Gahm's DTX role, Chandhoke's link to Roman AI, Merrill's link to Oncovera, Rizzotti's title, Lee Dunn's identity, everything about Josh Siegel / F-ADA. Corrected along the way: Fanelli was never DoD CIO; Booher's background is DARPA, Lockheed, Boeing, HSBC, Colgate, not Booz Allen; Sacks left the AI czar role in March 2026.

## 4. The numbers to open with (all sourced in docs/Verity\_Brief.md)

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- FY2025 federal improper payments: **\$186B** (GAO, April 2026). Medicare FFS \$28.83B, Medicare Advantage \$23.67B, Medicaid \$37.39B (CMS, January 2026).
- DOJ June 2026 National Health Care Fraud Takedown: 455 defendants, **\$6.5B**, 1,079 providers suspended, 1,403 billing privileges revoked.
- CMS six-month nationwide moratorium on new hospices and home health agencies (May 2026); ~800 LA hospices and HHAs suspended; Medicaid Fraud War Room stopped \$203M in its first 88 days.
- 2.8M people enrolled in Medicaid or ACA plans in multiple states (~\$14B/yr); OIG: 40 states paid \$50.3M to 584 terminated providers.
- Skin substitutes: \$256M (2019) to over \$10B (2024) in Part B; CMS projects \$19B in 2026 savings from the new flat rate.
- Minnesota's Medicaid director complained CMS "has not provided data or explanation on how the deferral amount was calculated." Verity is the transparent, reproducible version of the CMS war room.

## 5. The data (135 GB, 268 files, all real, all public)

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Everything lives in ~/Documents/Projects/verity/data/, each file with a .done marker and its source URL in manifest.json and INVENTORY.txt. The downloader (scripts/download\_all.py) is resumable.

| Folder                                                  | Contents                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| medicaid_tmsis                                          | HHS Open Data: Medicaid provider spending by HCPCS (3.07 GB parquet, 238,015,729 rows: billing NPI, servicing NPI, HCPCS, month, patients, claim lines, paid; no state column, join to enrollment on NPI), spending by NDC (21.5M rows), provider enrollment segments (66M rows, has state), Medicare revoked mirror. |
| cms_enrollment                                          | Hospice, HHA, SNF, hospital, FQHC, RHC enrollments and All Owners; SNF and hospital change of ownership; PECOS public enrollment (2.98M rows); Order and Referring; revalidation lists; opt-outs; Revoked Providers (8,136 rows); nursing home chains.                                                                |
| cms_program_integrity,<br>cms_pos                       | Market Saturation and Utilization by state-county and CBSA (with moratorium flag); Provider of Services files; CLIA labs.                                                                                                                                                                                             |
| cms_utilization                                         | All years 2013 to 2024: physician by provider and by provider-and-service, Part D by provider and by provider-and-drug, DME by referring provider and by supplier, inpatient and outpatient by provider-and-service, post-acute PUFs for hospice, HHA, SNF, IRF, LTCH.                                                |
| nppes, oig_leie,<br>sam_exclusions,<br>state_exclusions | NPPES full August 2026 file (9.7M NPIs) plus deactivated report; OIG LEIE (83,842 rows); SAM.gov public exclusions extract (168,304 rows, 7,201 with NPI, incl. 2,447 OPM/FEHB debarments); California, New York and Texas Medicaid exclusion lists.                                                                  |
| open_payments, openfda,<br>care_compare                 | Open Payments 2024 and 2025 general payments, 2025 ownership, all-years summaries; openFDA shortages, recalls, NDC directory; Care Compare hospice, home health, hospital, nursing home, LTCH, IRF, HHVBP, CAHPS, national clinician file.                                                                            |

## 6. Repository state

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ingest/ 00_clean_encoding.py, 01_build_warehouse.sql + build_warehouse.py,
02_timecodes.csv (318 codes), 03_sanity_checks.py, 04_build_monthly_and_aux.sql,
05_mn_rates_and_state_exclusions.py, 06_sam_exclusions.py, supabase_schema.sql,
sync_supabase.py, upload_storage.py, export_derived.py
detectors/ d3_revoked_but_paid.py (events table, paid-after join, cross-state test,
NPI check-digit self-test)
api/ main.py (FastAPI), verify.py (NPI and claims verification, SAM live lookup),
bluebutton.py (OAuth + PKCE client, EOB normalization)
web/ Next.js app wired to Supabase
docs/ Verity_Brief.md (judge dossier, idea, demo script, 24-hour plan), DATA.md
(data dictionary and join keys), methods.md, SUPABASE.md
data/verity.duckdb the warehouse: spend (56.7M rows of time-based Medicaid billing,
$532.8B), spend_totals, enroll, provider_state, hospice/hha/snf/hospital, owners
(613,574 rows), chow, ppef, order_referring, optout, revoked, leie, sam,
state_exclusions, saturation*, npes, timecodes, d3_* tables
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**API routes** (uvicorn api.main:app --port 8000): /health, /verify/npi/{npi}, POST /verify/claims, /sam/lookup?name=, /bluebutton/login, /bluebutton/callback, /bluebutton/me, /bluebutton/claims, /bluebutton/verify, POST /bluebutton/verify-bundle. Verified live: NPI 1811937436 returns a real 2026 OIG exclusion, SAM HHS prohibition and California Medi-Cal suspension; NPI 1801839063 returns a real 2025 CMS revocation under 424.535(a)(2) plus its LEIE entry.

## 7. Keys and integrations

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**SAM.gov** (SAM\_API\_KEY in .env): the daily public exclusions extract is pulled through the Data Services API and loaded as the sam table; the Exclusions API v4 is used live for name and UEI lookups. Both tested and working.

**Blue Button 2.0** (BLUEBUTTON\_\* in .env): sandbox app "Verity" is registered and its client ID and secret are accepted by CMS's token endpoint. The scopes line in .env was wrong and is fixed. Per your instruction that nothing be synthetic, all sample bundles and synthetic test users were removed, the base URL points at the production API, and /bluebutton/login refuses to run against the sandbox unless BLUEBUTTON\_ALLOW\_SANDBOX=1. Production access requires CMS approval through the Production Access Guide and cannot be obtained before Sunday. The real-data path that works today is a beneficiary's own Medicare.gov claims export through POST /bluebutton/verify-bundle or scripts/bluebutton\_offline\_verify.py. Recommendation: keep Blue Button out of the live demo, or present it as built and pending CMS approval.

Nothing in the repo is synthetic or demo data. Every input is public federal data or a real government API. There is no PII or PHI.

## 8. Demo script (5 minutes)

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Open cold: "Medicaid and Medicare lost \$66 billion to improper payments last year and the government just released the data to find it. We built Verity." Then the national risk map; one Los Angeles hospice cluster in the ownership graph with an owner on the exclusion list; one Minnesota provider billing more hours than exist in a

day; a revoked-in-Medicare provider still paid by Medicaid; the agent drafting the referral packet live; the reviewer approving and the score updating. Close on three numbers: dollars flagged, providers flagged, minutes per referral versus days for a retrospective audit. Last slide: who buys this Monday.

**Commercial story:** every state Medicaid agency must implement cross-state duplicate checks, mortality matching, address verification and provider revalidation under OBBBA by December 31, 2026, and the Task Force can withhold federal funds from states that fail. Fifty buyers, one deadline. Then Medicare Advantage plans and MCOs who eat the loss when they pay a ghost network. Competitors: Palantir Foundry at CMS (CMS only, at CMS prices), SAS and Codoxo for payers; nobody is open, public-data-first and state-ready for the deadline.

**Pre-empted objections:** Verity never touches a beneficiary or denies a claim; it scores providers and networks, and a human signs every action. Evaluation for the ML judge: precision at K against held-out revocations with a time split (features 2018 to 2022, revocations 2023 to 2024). Say what it can't catch: collusive beneficiaries, Medicare Advantage diagnosis upcoding. Humility scores with Fanelli.

## 9. Open items

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- Detectors 1 (ghost networks) and 2 (impossible days) are specified in the brief and the warehouse is built for them; Detector 3 has code. Get one real cluster and one real impossible-day case on screen before anything else.
- The Next.js console (overview, clusters, flags, provider, Minnesota, methods pages) is built and wired to Supabase; the packet panel and review loop are in; the serving tables fill once D1 and sync\_outputs run.
- Housekeeping on your machine: trash \_to\_delete/ (3 GB of parquet download chunks plus the removed synthetic Blue Button files). api/\_\_pycache\_\_/ is gitignored.
- The git repo has no commits yet; commit before the deadline.